

HYDROGENATED NBR SYNTHESIS AND PROCESS DEVELOPMENT ACROSS SELECTIVE HYDROGENATION AND METATHESIS ROUTES

1. ABSTRACT

This report synthesizes seven primary patents relevant to hydrogenated nitrile rubber (HNBR) preparation, with emphasis on selective hydrogenation of NBR, low-molecular-weight HNBR production, and catalyst systems that combine metathesis with hydrogenation. The target material is hydrogenated or partially hydrogenated nitrile rubber derived from acrylonitrile-butadiene rubber, with the desired attributes of improved heat, ozone, and oil resistance, low residual unsaturation, and controlled molecular weight/Mooney viscosity. The patent landscape separates into two major process families. First, conventional solution hydrogenation of NBR is disclosed using ruthenium, palladium, or rhodium catalysts under elevated temperature and hydrogen pressure, with attention to catalyst loading, residual double bond reduction, and catalyst-removal issues. Second, several patents address a more integrated route in which metathesis and hydrogenation occur simultaneously or sequentially using Ru/Os carbene catalysts, often with co-olefins and catalyst pretreatment, to reduce molecular weight while hydrogenating the polymer backbone. Across the selected patents, the key formulation/process trends are: use of organic solvent solution processing; hydrogen pressures from 4.137 MPa to 15,000,000 Pa; temperatures commonly around 120 °C or within 60–200 °C; catalyst loadings reported in phr or ppm; and explicit control of residual double bond content, Mooney viscosity, and polydispersity. The metathesis-based patents emphasize low-Mw HNBR with narrower molecular weight distributions and improved processability, while the catalyst-composition patent highlights pretreatment of Ru catalysts with vinyl ethyl ether to improve hydrogenation efficiency and preserve molecular weight. The more recent EP patent focuses on reducing catalyst residues and improving ageing properties by using low noble-metal levels and avoiding post-hydrogenation removal steps. Collectively, these disclosures show that HNBR synthesis is moving toward more selective, lower-residue, and process-integrated hydrogenation strategies that balance high conversion with molecular-weight retention and downstream performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is directed to nitrile rubbers and their work-up, not to hydrogenation. It discloses emulsion polymerized NBR latices and emphasizes coagulation/washing conditions that influence calcium, chlorine, and thio-end-group content, which in turn affect storage stability and processability. The patent discusses selective hydrogenation only as prior art rather than as the invention itself. - Acrylonitrile content: 38.6% by weight — Example 36 - Coagulation salt: CaCl_2 — Example 36 - Salt concentration: 0.3% by weight — Example 36 - Salt amount based on NBR: 1.8% by weight — Example 36 - Coagulation temperature: 90 °C — Example 36 - Washing temperature: 60 °C — Example 36 - Washing time: 8 h — Example 36 - Chlorine content: 100 ppm — Example 36 - Calcium content: 1930 ppm — Example 36 - ML1+4: 47 MU — Example 36 - MV2: 51 MU — Example 36 - Storage stability: 4 MU — Example 36 - Calcium content: 150 ppm to 1290 ppm — Claim 1 - Chlorine content: at least 40 ppm — Claim 1 - Storage stability: not more than 3 Mooney units — Claim 1

Relevant Experimental Evidence - Example 36: NBR latex coagulated with CaCl_2 and washed with calcium-ion-containing water showed specific calcium/chlorine retention and storage stability data.

Technical Relevance This patent is only indirectly relevant because it addresses NBR precursor quality and work-up, which can affect later hydrogenation behavior. It does not disclose HNBR synthesis, but it provides context on how latex isolation conditions influence polymer properties important for HNBR feedstocks.

Patent 2

- Patent Number: US10500578B2
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber in solution using Ru/Os-based catalyst compositions formed by contacting a complex catalyst with a co-catalyst such as vinyl ethyl ether. The catalyst pretreatment improves hydrogenation activity and helps preserve molecular weight during HNBR formation; the patent also includes an optional metathesis step before hydrogenation. - Perbunan® 3431VP concentration: 6 wt % — Example 1 - MCB amount: 282 g — Example 1 - Nitrogen bubbling time: 30 minutes — Example 1 - Reaction temperature: 120 °C — Example 1 - Wilkinson's catalyst: 15 mg — Example 1 - PPh₃: 18 mg — Example 1 - Hydrogen pressure: 4.137 MPa — Example 1 - Agitation speed: 800 rpm — Example 1 - Hydrogenation time: 5 h — Example 1 - Hydrogenation degree: 90.3% — Example 1 - Mn: 76,286 g/mol — Example 1 - Mw: 260,572 g/mol — Example 1 - PDI: 3.42 — Example 1 - Perbunan® 3431VP concentration: 3 wt % — Example 2 - Catalyst (2): 9 mg — Example 2 - Hydrogenation time: 4 h — Example 2 - Hydrogenation degree: 98.6% — Example 2 - Mn: 5,560 g/mol — Example 2 - Mw: 14,407 g/mol — Example 2 - PDI: 2.59 — Example 2 - Catalyst (2): 18 mg — Example 3 - Hydrogenation time: 18 h — Example 3 - Hydrogenation degree: 92.6% — Example 3 - Mn: 10,103 g/mol — Example 3 - Mw: 19,964 g/mol — Example 3 - PDI: 1.98 — Example 3 - Vinyl ethyl ether pretreatment time: 12 h — Example 4 - Catalyst (2): 18 mg — Example 4 - Hydrogenation time: 2 h — Example 4 - Hydrogenation degree: 99% — Example 4 - Mn: 74,495 g/mol — Example 4 - Mw: 229,568 g/mol — Example 4 - PDI: 3.08 — Example 4 - Vinyl ethyl ether amount: 100 µL — Example 5 - Catalyst (2): 9 mg — Example 5 - Hydrogenation time: 4 h — Example 5 - Hydrogenation degree: 95% — Example 5 - Mn: 71,220 g/mol — Example 5 - Mw: 224,342 g/mol — Example 5 - PDI: 3.15 — Example 5 - Hydrogenation degree: 96% — Example 6 - Mn: 78,125 g/mol — Example 6 - Mw: 256,043 g/mol — Example 6 - PDI: 3.28 — Example 6 - Hydrogenation degree: 99% — Example 7 - Mn: 78,514 g/mol — Example 7 - Mw: 222,813 g/mol — Example 7 - PDI: 2.84 — Example 7 - Hydrogen pressure: 6.895 MPa — Example 8 - Catalyst (2): 18 mg — Example 8 - Hydrogenation time: 3 h — Example 8 - Hydrogenation degree: 98% — Example 8 - Mn: 107,058 g/mol — Example 8 - Mw: 261,844 g/mol — Example 8 - PDI: 2.44 — Example 8 - Catalyst (1): 9 mg — Example 9 - Hydrogenation time: 3 h — Example 9 - Hydrogenation degree: 97% — Example 9 - Mn: 77,473 g/mol — Example 9 - Mw: 219,498 g/mol — Example 9 - PDI: 2.83 — Example 9 - Hydrogenation temperature: 120 °C — Table 2 - Hydrogenation pressure: 4.137 MPa — Table 2 - Hydrogenation pressure: 6.895 MPa — Table 2 - Catalyst amount: 0.0246 mmol — Table 2 - Co-catalyst amount: 42.4 mmol — Table 2 - Metathesis time: 12 h — Table 2 - Hydrogenation time: 2 h to 4 h — Table 2 - Hydrogenation degree: 95% to 99% — Table 2 - Metathesis time: 1 h — Example 10 - VEE amount: 0.5 mL — Example 10 - Stirring time after VEE addition: 10 minutes — Example 10 - Hydrogenation time: 3 h — Example 10 - Hydrogenation degree: 99% — Example 10 - Metathesis time: 2 h — Example 11 - Hydrogenation time: 3 h — Example 11 - Hydrogenation degree: 99% — Example 11 - Hydrogenation pressure: 4.137 MPa — Table 3 - Hydrogenation temperature: 120 °C — Table 3 - Hydrogenation time: 3 h — Table 3

Relevant Experimental Evidence - Example 1: Baseline hydrogenation with Wilkinson's catalyst reached 90.3% hydrogenation in 5 h but retained relatively high Mw and PDI. - Example 2: Catalyst (2) without pretreatment achieved 98.6% hydrogenation in 4 h but caused severe molecular-weight loss. - Example 4: VEE-pretreated Catalyst (2) gave 99% hydrogenation in only 2 h while preserving much higher molecular weight. - Example 8: Higher hydrogen pressure (6.895

MPa) with Catalyst (2) still delivered 98% hydrogenation and moderate molecular-weight retention. - Example 10: In-situ metathesis followed by hydrogenation produced 99% hydrogenation with low PDI and reduced Mw.

Technical Relevance This is a core HNBR synthesis patent because it directly teaches selective hydrogenation of NBR with catalyst pretreatment to improve activity and control polymer degradation. It is especially relevant for balancing high conversion, low residual unsaturation, and retention of useful molecular weight.

Patent 3

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route in which NBR is first subjected to olefin metathesis in the presence of Ru or Os catalysts and optionally a co-olefin, then hydrogenated to form low-molecular-weight HNBR. The process is designed to reduce molecular weight and polydispersity while maintaining low residual double bond content and acceptable Mooney viscosity. - MW: 30,000 to 250,000 — Claim 1 - Mooney viscosity: less than 10 — Claim 1 - PDI: less than 2.5 — Claim 1 - MW: 40,000 to 220,000 — Claim 2 - PDI: less than 2.3 — Claim 3 - 200 g rubber — Example 1 - 1133 g MCB — Example 1 - 15 wt.-% solid — Example 1 - 6.9×10^5 Pa C₂H₄ — Example 1 - 200 g rubber — Example 2 - 1133 g MCB — Example 2 - 15 wt.-% solid — Example 2 - 6.9×10^5 Pa C₂H₄ — Example 2 - 75 g rubber — Example 3 - 1175 g MCB — Example 3 - 6 wt.-% solid — Example 3 - 6.9×10^5 Pa C₂H₄ — Example 3 - 75 g rubber — Example 4 - 1175 g MCB — Example 4 - 150 g 1-hexene — Example 4 - 80 °C — Table 1 - 500 psi co-olefin concentration — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 3 — Table 2 - Mn: 73,000 g/mol — Table 2 - Mw: 189,000 g/mol — Table 2 - PDI: 2.59 — Table 2 - Mooney viscosity: 43 — Table 2 - Mn: 60,000 g/mol — Table 2 - Mw: 136,000 g/mol — Table 2 - PDI: 2.27 — Table 2 - Mooney viscosity: 28 — Table 2 - Mn: 31,000 g/mol — Table 2 - Mw: 59,000 g/mol — Table 2 - PDI: 1.90 — Table 2 - Mooney viscosity: 3 — Table 2 - Mn: 55,000 g/mol — Table 2 - Mw: 111,000 g/mol — Table 2 - PDI: 2.02 — Table 2 - Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis reduced Mw and Mooney viscosity relative to the starting polymer. - Example 2: Higher catalyst loading further reduced Mw and Mooney viscosity. - Example 3: Lower solids and ethylene metathesis produced the lowest PDI and very low Mooney viscosity. - Example 4: 1-hexene co-olefin gave low-Mw

HNBR with Mooney viscosity of 31. - Example 4: The table shows up to a fivefold reduction in Mw and PDI as low as 1.90.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR via metathesis followed by hydrogenation. It provides concrete process windows for achieving low Mooney viscosity and narrow molecular-weight distribution, both important for processability.

Patent 4

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the preparation of low-molecular-weight HNBR by olefin metathesis of NBR using co-olefins such as ethylene or 1-hexene, followed by hydrogenation. The process is aimed at reducing molecular weight and Mooney viscosity while maintaining a hydrogenated nitrile rubber structure suitable for shaped articles. - MW: 30,000 to 250,000 — Claim 1 - Mooney viscosity: less than 10 — Claim 1 - PDI: less than 2.5 — Claim 1 - MW: 40,000 to 220,000 — Claim 2 - PDI: less than 2.3 — Claim 3 - 200 g rubber — Example 1 - 1133 g MCB — Example 1 - 15 wt.-% solid — Example 1 - 6.9×10^5 Pa C₂H₄ — Example 1 - 200 g rubber — Example 2 - 1133 g MCB — Example 2 - 15 wt.-% solid — Example 2 - 6.9×10^5 Pa C₂H₄ — Example 2 - 75 g rubber — Example 3 - 1175 g MCB — Example 3 - 6 wt.-% solid — Example 3 - 6.9×10^5 Pa C₂H₄ — Example 3 - 75 g rubber — Example 4 - 1175 g MCB — Example 4 - 150 g 1-hexene — Example 4 - 80 °C — Table 1 - 500 psi co-olefin concentration — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mooney viscosity: 3 — Table 2 - Mn: 73,000 g/mol — Table 2 - Mw: 189,000 g/mol — Table 2 - PDI: 2.59 — Table 2 - Mooney viscosity: 43 — Table 2 - Mn: 60,000 g/mol — Table 2 - Mw: 136,000 g/mol — Table 2 - PDI: 2.27 — Table 2 - Mooney viscosity: 28 — Table 2 - Mn: 31,000 g/mol — Table 2 - Mw: 59,000 g/mol — Table 2 - PDI: 1.90 — Table 2 - Mooney viscosity: 3 — Table 2 - Mn: 55,000 g/mol — Table 2 - Mw: 111,000 g/mol — Table 2 - PDI: 2.02 — Table 2 - Mooney viscosity: 31 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis reduced Mw and Mooney viscosity relative to the starting polymer. - Example 2: Higher catalyst loading further reduced Mw and Mooney viscosity. - Example 3: Lower solids and ethylene metathesis produced the lowest PDI and very low Mooney viscosity. - Example 4: 1-hexene co-olefin gave low-Mw HNBR with Mooney viscosity of 31. - Example 4: The table shows up to a fivefold reduction in Mw

and PDI as low as 1.90.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR via metathesis followed by hydrogenation. It provides concrete process windows for achieving low Mooney viscosity and narrow molecular-weight distribution, both important for processability.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches simultaneous metathesis and hydrogenation of NBR in the presence of hydrogen using Ru or Os carbene catalysts, optionally with a co-olefin. The process is positioned as a single-step route to low-molecular-weight HNBR with reduced residual double bond content and improved processability compared with conventional sequential methods. - Hydrogenation conversion: higher than 95% — synthesis discussion - Residual double bond content: <5% — synthesis discussion - Gel level: less than about 2.5% — synthesis discussion - Hydrogenation degree: 50 to 100% — synthesis discussion - Hydrogenation degree: about 80 to about 100% — synthesis discussion - Hydrogenation degree: about 90 to about 99.9% — synthesis discussion - Remaining unsaturation: below 18% — synthesis discussion - Acrylonitrile content: roughly up to about 50% — synthesis discussion - Mooney viscosity: below 55 — synthesis discussion - Mw: 30,000 to 250,000 — claim context - Mooney viscosity: 3 to 50 — claim context - PDI: less than 2.5 — claim context - Mooney viscosity: less than 10 — claim 1 - Mw: 40,000 to 220,000 — claim 2 - PDI: less than 2.3 — claim 3 - Hydrogenation pressure: 4.137 MPa — Example 1 context - Hydrogenation temperature: 120 °C — Example 1 context - Hydrogenation time: 3 h — Example 1 context - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 0 to 240 min — Table 2 - RDB: 100% to 0.4% — Table 2 - RDB: 100% to 3.5% — Table 2 - RDB: 100% to 0.7% — Table 2 - Mw: 113,371 g/mol — Table 3 - Mn: 48,352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133,758 g/mol — Table 3 - Mn: 54,595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195,304 g/mol — Table 3 - Mn: 64,130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave 0.4% RDB after 240 min with Mw 113,371 g/mol and Mooney 33.3. - Example 1: Arlt catalyst gave 3.5% RDB after 240 min

with Mw 133,758 g/mol and Mooney 41.3. - Example 1: Wilkinson's catalyst gave 0.7% RDB after 240 min but the highest Mw and Mooney viscosity. - Example 1: The inventive catalyst outperformed the industrial norm in hydrogenation efficiency. - Example 1: The data show that catalyst choice strongly affects residual unsaturation and molecular-weight retention.

Technical Relevance This patent is directly relevant because it discloses a one-step simultaneous hydrogenation/metathesis route to HNBR. It is especially important for low-residual-unsaturation HNBR with controlled molecular weight and reduced process complexity.

Patent 6

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of NBR in the presence of hydrogen using Ru or Os compounds, optionally with a co-olefin, to produce low-molecular-weight HNBR in a single step. The process emphasizes catalyst selection and catalyst pretreatment to improve hydrogenation efficiency while limiting molecular-weight loss. - Hydrogenation conversion: higher than 95% — synthesis discussion - Residual double bond content: <5% — synthesis discussion - Gel level: less than about 2.5% — synthesis discussion - Hydrogenation degree: 50 to 100% — synthesis discussion - Hydrogenation degree: about 80 to about 100% — synthesis discussion - Hydrogenation degree: about 90 to about 99.9% — synthesis discussion - Remaining unsaturation: below 18% — synthesis discussion - Acrylonitrile content: roughly up to about 50% — synthesis discussion - Mooney viscosity: below 55 — synthesis discussion - Mw: 20,000 to 250,000 — claim context - Mooney viscosity: 1 to 50 — claim context - PDI: less than 3.0 — claim context - Hydrogenation pressure: 4.137 MPa — Example 1 context - Hydrogenation temperature: 120 °C — Example 1 context - Hydrogenation time: 240 min — Example 1 context - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 0 to 240 min — Table 2 - RDB: 100% to 0.4% — Table 2 - RDB: 100% to 3.5% — Table 2 - RDB: 100% to 0.7% — Table 2 - Mw: 113,371 g/mol — Table 3 - Mn: 48,352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133,758 g/mol — Table 3 - Mn: 54,595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195,304 g/mol — Table 3 - Mn: 64,130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave 0.4% RDB after 240 min with Mw 113,371 g/mol and Mooney 33.3. - Example 1: Arlt catalyst gave 3.5% RDB after 240 min

with Mw 133,758 g/mol and Mooney 41.3. - Example 1: Wilkinson's catalyst gave 0.7% RDB after 240 min but the highest Mw and Mooney viscosity. - Example 1: The inventive catalyst outperformed the industrial norm in hydrogenation efficiency. - Example 1: The data show that catalyst choice strongly affects residual unsaturation and molecular-weight retention.

Technical Relevance This patent is directly relevant because it discloses a one-step simultaneous hydrogenation/metathesis route to HNBR. It is especially important for low-residual-unsaturation HNBR with controlled molecular weight and reduced process complexity.

Patent 7

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of NBR in solution using ruthenium, palladium, or rhodium compounds, with a strong focus on catalyst residue management and ageing performance of the resulting HNBR. The process is designed to produce HNBR with good ageing properties while minimizing the need for elaborate post-hydrogenation catalyst removal. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200 °C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Conjugated diene content: 40% to 90% by weight — Claim 1 - Unsaturated nitrile content: 10% to 60% by weight — Claim 1 - Gel content increase: less than 10% during 4 days at 140 °C — synthesis discussion - Mooney viscosity increase: less than 40% during 4 days at 140 °C — synthesis discussion - Rhodium content: 58 to 60 ppm — prior-art discussion - Pd catalyst amount: 0.45 parts — prior-art discussion - RDB: 5.8 to 9.2 ppm Rh in purified HNBR — prior-art discussion

Relevant Experimental Evidence - No parser-detected examples were provided; the patent text instead emphasizes process conditions and catalyst-removal strategies.

Technical Relevance This patent is relevant because it addresses practical HNBR production with controlled noble-metal catalyst levels and improved ageing properties. It complements the

synthesis-focused patents by emphasizing catalyst residue control, a key industrial issue for HNBR quality.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10500578B2 and EP3827029A1 both use solution-phase hydrogenation of NBR, but US10500578B2 emphasizes Ru catalyst pretreatment with vinyl ethyl ether and simultaneous metathesis/hydrogenation, whereas EP3827029A1 focuses on low ppm noble-metal loading and ageing/catalyst-residue control.
- US7919563B2, US7772328B2, EP2473281B1, and US9150669B2 all use metathesis-assisted molecular-weight reduction, but EP2473281B1 and US9150669B2 uniquely combine metathesis and hydrogenation in a single step under hydrogen, while US7919563B2 and US7772328B2 describe metathesis followed by hydrogenation.
- The metathesis-based patents report low Mooney viscosities and reduced Mw: US7919563B2 and US7772328B2 show Mooney values as low as 3 and PDI down to 1.90, while EP2473281B1 and US9150669B2 target Mw 20,000 to 250,000 and PDI below 3.0 with residual double bond content below 5%.
- US10500578B2 uses hydrogen pressures of 4.137 MPa and 6.895 MPa at 120 °C, whereas EP3827029A1 claims a broader hydrogenation window of 60 to 200 °C and 700000 to 15000000 Pa.
- Catalyst systems differ substantially: US10500578B2 centers on Ru/Os carbene complexes with co-catalyst pretreatment; EP2473281B1 and US9150669B2 use formula (I) Ru/Os metathesis/hydrogenation catalysts; EP3827029A1 uses Ru, Pd, or Rh compounds for conventional hydrogenation; US7919563B2 and US7772328B2 rely on Ru/Os metathesis catalysts plus co-olefins such as ethylene or 1-hexene.

4. CONCLUSION

The evidence shows two viable HNBR synthesis strategies: conventional solution hydrogenation of NBR with noble-metal catalysts, and metathesis-assisted molecular-weight reduction followed by or concurrent with hydrogenation. For high-performance HNBR, the strongest evidence supports low residual double bond content below 5%, hydrogenation temperatures around 120 °C, and hydrogen pressures in the 4.137 to 6.895 MPa range for the catalyst-composition route, or 60 to 200 °C and 700000 to 15000000 Pa for broader hydrogenation practice. For low-Mw, highly processable HNBR, the metathesis-based patents recommend Ru/Os catalysts with co-olefins and, in the simultaneous route, hydrogen present during metathesis to achieve Mw 20,000 to 250,000, Mooney viscosity below 10 to 50, and PDI below 2.5 to 3.0. Catalyst pretreatment with vinyl ethyl ether is a clear performance enhancer, improving hydrogenation rate and helping preserve molecular weight. Overall, the best-supported synthesis parameters are selective hydrogenation in solution, controlled noble-metal loading, and either sequential or simultaneous

metathesis/hydrogenation depending on whether the priority is maximum unsaturation removal or molecular-weight reduction.

5. REFERENCES

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- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
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- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>